4 Describing Data:

DISPLAYING AND EXPLORING DATA

MCGIVERN JEWELERS recently posted an advertisement on a social media site reporting the shape, size, price, and cut grade for 33 of its diamonds in stock. Develop a box plot of the variable price and comment on the result. (See Exercise 37 and LO4-4.)

LEARNING OBJECTIVES

When you have completed this chapter, you will be able to:

LO4-1 Construct and interpret a dot plot.

LO4-2 Construct and describe a stem-and-leaf display.

LO4-3 Identify and compute measures of position.

LO4-4 Construct and analyze a box plot.

LO4-5 Compute and interpret the coefficient of skewness.

LO4-6 Create and interpret a scatter diagram.

LO4-7 Develop and explain a contingency table.

INTRODUCTION

Chapter 2 began our study of descriptive statistics. In order to transform raw or ungrouped data into a meaningful form, we organize the data into a frequency distribution. We present the frequency distribution in graphic form as a histogram or a frequency polygon. This allows us to visualize where the data tend to cluster, the largest and the smallest values, and the general shape of the data.

In Chapter 3, we first computed several measures of location, such as the mean, median, and mode. These measures of location allow us to report a typical value in the set of observations. We also computed several measures of dispersion, such as the range, variance, and standard deviation. These measures of dispersion allow us to describe the variation or the spread in a set of observations.

We continue our study of descriptive statistics in this chapter. We study (1) dot plots, (2) stem-and-leaf displays, (3) percentiles, and (4) box plots. These charts and statistics give us additional insight into where the values are concentrated as well as the general shape of the data. Then we consider bivariate data. In bivariate data, we observe two variables for each individual or observation. Examples include the number of hours a student studied and the points earned on an examination; if a sampled product meets quality specifications and the shift on which it is manufactured; or the amount of electricity used in a month by a homeowner and the mean daily high temperature in the region for the month. These charts and graphs provide useful insights as we use business analytics to enhance our understanding of data.

LO4-1
Construct and interpret a dot plot.

DOT PLOTS

Recall for the Applewood Auto Group data, we summarized the profit earned on the 180 vehicles sold with a frequency distribution using eight classes. When we organized the data into the eight classes, we lost the exact value of the observations. A **dot plot,** on the other hand, groups the data as little as possible, and we do not lose the identity of an individual observation. To develop a dot plot, we display a dot for each observation along a horizontal number line indicating the possible values of the data. If there are identical observations or the observations are too close to be shown individually, the dots are "piled" on top of each other. This allows us to see the shape of the distribution, the value about which the data tend to cluster, and the largest and smallest observations. Dot plots are most useful for smaller data sets, whereas histograms tend to be most useful for large data sets. An example will show how to construct and interpret dot plots.

EXAMPLE

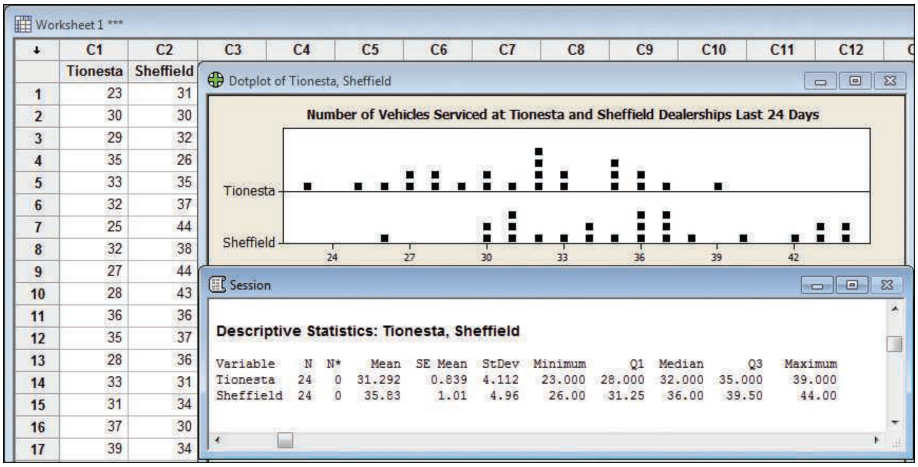
The service departments at Tionesta Ford Lincoln and Sheffield Motors Inc., two of the four Applewood Auto Group dealerships, were both open 24 days last month. Listed below is the number of vehicles serviced last month at the two dealerships. Construct dot plots and report summary statistics to compare the two dealerships.

Tionesta Ford Lincoln					
Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
23	33	27	28	39	26
30	32	28	33	35	32
29	25	36	31	32	27
35	32	35	37	36	30

Sheffield Motors Inc.					
Monday	**Tuesday**	**Wednesday**	**Thursday**	**Friday**	**Saturday**
31	35	44	36	34	37
30	37	43	31	40	31
32	44	36	34	43	36
26	38	37	30	42	33

SOLUTION

The Minitab system provides a dot plot and outputs the mean, median, maximum, and minimum values, and the standard deviation for the number of cars serviced at each dealership over the last 24 working days.



The dot plots, shown in the center of the output, graphically illustrate the distributions for each dealership. The plots show the difference in the location and dispersion of the observations. By looking at the dot plots, we can see that the number of vehicles serviced at the Sheffield dealership is more widely dispersed and has a larger mean than at the Tionesta dealership. Several other features of the number of vehicles serviced are:

- Tionesta serviced the fewest cars in any day, 23.
- Sheffield serviced 26 cars during their slowest day, which is 4 cars less than the next lowest day.
- Tionesta serviced exactly 32 cars on four different days.
- The numbers of cars serviced cluster around 36 for Sheffield and 32 for Tionesta.

From the descriptive statistics, we see Sheffield serviced a mean of 35.83 vehicles per day. Tionesta serviced a mean of 31.292 vehicles per day during the same period. So Sheffield typically services 4.54 more vehicles per day. There is also more dispersion, or variation, in the daily number of vehicles serviced at Sheffield than at Tionesta. How do we know this? The standard deviation is larger at Sheffield (4.96 vehicles per day) than at Tionesta (4.112 cars per day).

LO4-2
Construct and describe a stem-and-leaf display.

STEM-AND-LEAF DISPLAYS

In Chapter 2, we showed how to organize data into a frequency distribution so we could summarize the raw data into a meaningful form. The major advantage to organizing the data into a frequency distribution is we get a quick visual picture of the shape of the distribution without doing any further calculation. To put it another way, we can see where the data are concentrated and also determine whether there are any extremely large or small values. There are two disadvantages, however, to organizing the data into a frequency distribution: (1) we lose the exact identity of each value and (2) we are not sure how the values within each class are distributed. To explain, the Theater of the Republic in Erie, Pennsylvania, books live theater and musical performances. The theater's capacity is 160 seats. Last year, among the forty-five performances, there were eight different plays and twelve different bands. The following frequency distribution shows that between eighty up to ninety people attended two of the forty-five performances; there were seven performances where ninety up to one hundred people attended. However, is the attendance within this class clustered about 90, spread evenly throughout the class, or clustered near 99? We cannot tell.

Attendance	Frequency
80 up to 90	2
90 up to 100	7
100 up to 110	6
110 up to 120	9
120 up to 130	8
130 up to 140	7
140 up to 150	3
150 up to 160	3
Total	45

One technique used to display quantitative information in a condensed form and provide more information than the frequency distribution is the **stem-and-leaf display.** An advantage of the stem-and-leaf display over a frequency distribution is we do not lose the identity of each observation. In the above example, we would not know the identity of the values in the 90 up to 100 class. To illustrate the construction of a stem-and-leaf display using the number people attending each performance, suppose the seven observations in the 90 up to 100 class are 96, 94, 93, 94, 95, 96, and 97. The **stem** value is the leading digit or digits, in this case 9. The **leaves** are the trailing digits. The stem is placed to the left of a vertical line and the leaf values to the right.

The values in the 90 up to 100 class would appear as follows:

9 | 6 4 3 4 5 6 7

It is also customary to sort the values within each stem from smallest to largest. Thus, the second row of the stem-and-leaf display would appear as follows:

9 | 3 4 4 5 6 6 7

With the stem-and-leaf display, we can quickly observe that 94 people attended two performances and the number attending ranged from 93 to 97. A stem-and-leaf display is similar to a frequency distribution with more information, that is, the identity of the observations is preserved.

STEM-AND-LEAF DISPLAY A statistical technique to present a set of data. Each numerical value is divided into two parts. The leading digit(s) becomes the stem and the trailing digit the leaf. The stems are located along the vertical axis, and the leaf values are stacked against each other along the horizontal axis.

The following example explains the details of developing a stem-and-leaf display.

EXAMPLE

Listed in Table 4–1 is the number of people attending each of the 45 performances at the Theater of the Republic last year. Organize the data into a stem-and-leaf display. Around what values does attendance tend to cluster? What is the smallest attendance? The largest attendance?

TABLE 4–1 Number of People Attending Each of the 45 Performances at the Theater of the Republic

96	93	88	117	127	95	113	96	108	94	148	156
139	142	94	107	125	155	155	103	112	127	117	120
112	135	132	111	125	104	106	139	134	119	97	89
118	136	125	143	120	103	113	124	138			

SOLUTION

From the data in Table 4–1, we note that the smallest attendance is 88. So we will make the first stem value 8. The largest attendance is 156, so we will have the stem values begin at 8 and continue to 15. The first number in Table 4–1 is 96, which has a stem value of 9 and a leaf value of 6. Moving across the top row, the second value is 93 and the third is 88. After the first 3 data values are considered, the chart is as follows.

Stem	Leaf
8	8
9	6 3
10	
11	
12	
13	
14	
15	

Organizing all the data, the stem-and-leaf chart looks as follows.

Stem	Leaf
8	8 9
9	6 3 5 6 4 4 7
10	8 7 3 4 6 3
11	7 3 2 7 2 1 9 8 3
12	7 5 7 0 5 5 0 4
13	9 5 2 9 4 6 8
14	8 2 3
15	6 5 5

The usual procedure is to sort the leaf values from the smallest to largest. The last line, the row referring to the values in the 150s, would appear as:

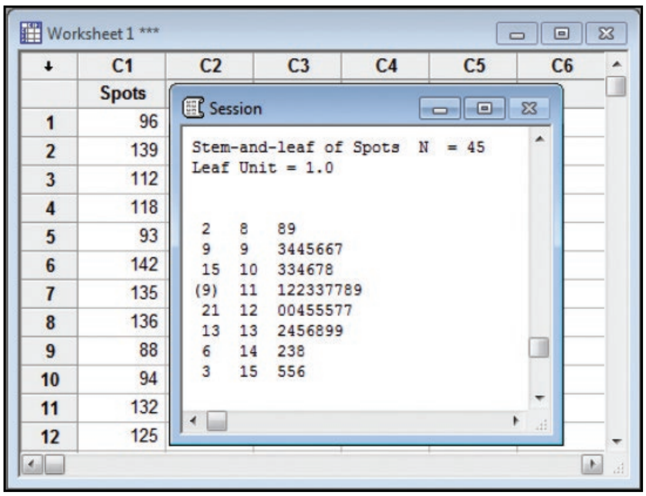
15	5	5	6

The final table would appear as follows, where we have sorted all of the leaf values.

Stem	Leaf
8	8 9
9	3 4 4 5 6 6 7
10	3 3 4 6 7 8
11	1 2 2 3 3 7 7 8 9
12	0 0 4 5 5 5 7 7
13	2 4 5 6 8 9 9
14	2 3 8
15	5 5 6

You can draw several conclusions from the stem-and-leaf display. First, the minimum number of people attending is 88 and the maximum is 156. There were two performances with less than 90 people attending, and three performances with 150 or more. You can observe, for example, that for the three performances with more than 150 people attending, the actual attendances were 155, 155, and 156. The concentration of attendance is between 110 and 130. There were fifteen performances with attendance between 110 and 119 and eight performances between 120 and 129. We can also tell that within the 120 to 129 group the actual attendances were spread evenly throughout the class. That is, 120 people attended two performances, 124 people attended one performance, 125 people attended three performances, and 127 people attended two performances.

We also can generate this information on the Minitab software system. We have named the variable *Attendance*. The Minitab output is below. You can find the Minitab commands that will produce this output in Appendix C.



The Minitab solution provides some additional information regarding cumulative totals. In the column to the left of the stem values are numbers such as 2, 9, 15, and so on. The number 9 indicates there are 9 observations that have occurred before the value of 100. The number 15 indicates that 15 observations have occurred prior to 110. About halfway down the column the number 9 appears in parentheses. The parentheses indicate that the middle value or median appears in that row and there are nine values in this group. In this case, we describe the middle value as the value below which half of the observations occur. There are a total of 45 observations, so the middle value, if the data were arranged from smallest to largest, would be the 23rd observation; its value is 118. After the median, the values begin to decline. These values represent the “more than” cumulative totals. There are 21 observations of 120 or more, 13 of 130 or more, and so on.

Which is the better choice, a dot plot or a stem-and-leaf chart? This is really a matter of personal choice and convenience. For presenting data, especially with a large number of observations, you will find dot plots are more frequently used. You will see dot plots in analytical literature, marketing reports, and occasionally in annual reports. If you are doing a quick analysis for yourself, stem-and-leaf tallies are handy and easy, particularly on a smaller set of data.

SELF-REVIEW 4–1

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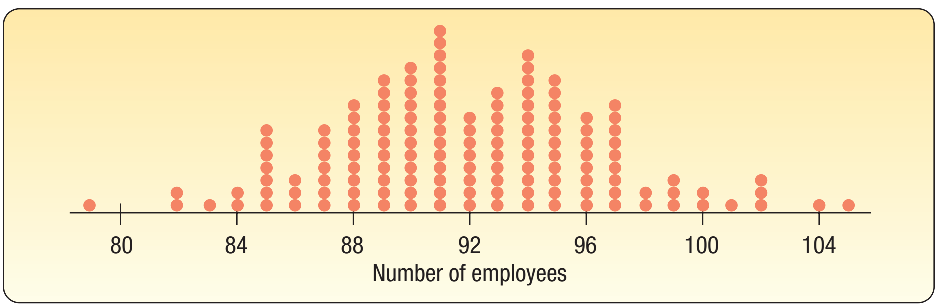
1. The number of employees at each of the 142 Home Depot stores in the Southeast region is shown in the following dot plot.



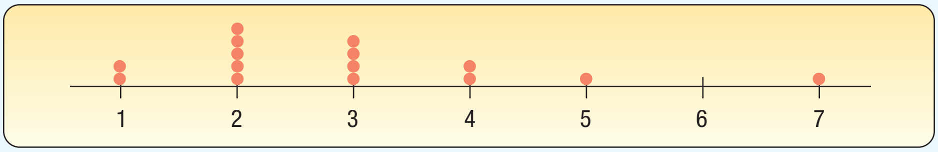
   (a) What are the maximum and minimum numbers of employees per store?
   (b) How many stores employ 91 people?
   (c) Around what values does the number of employees per store tend to cluster?
2. The rate of return for 21 stocks is:

8.3	9.6	9.5	9.1	8.8	11.2	7.7	10.1	9.9	10.8	
10.2	8.0	8.4	8.1	11.6	9.6	8.8	8.0	10.4	9.8	9.2

Organize this information into a stem-and-leaf display.
(a) How many rates are less than 9.0?
(b) List the rates in the 10.0 up to 11.0 category.
(c) What is the median?
(d) What are the maximum and the minimum rates of return?

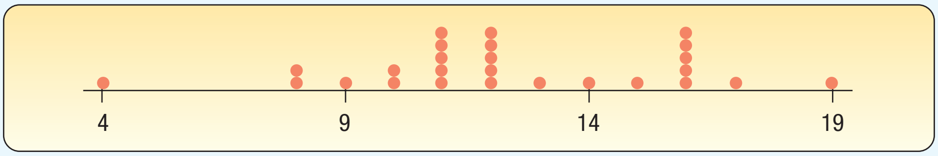
EXERCISES

1. Describe the differences between a histogram and a dot plot. When might a dot plot be better than a histogram?
2. Describe the differences between a histogram and a stem-and-leaf display.
3. Consider the following chart.



a. What is this chart called?
b. How many observations are in the study?
c. What are the maximum and the minimum values?
d. Around what values do the observations tend to cluster?

4. The following chart reports the number of cell phones sold at a big-box retail store for the last 26 days.



a. What are the maximum and the minimum numbers of cell phones sold in a day?
b. What is a typical number of cell phones sold?

5. The first row of a stem-and-leaf chart appears as follows: 62 | 1 3 3 7 9. Assume whole number values.
a. What is the "possible range" of the values in this row?
b. How many data values are in this row?
c. List the actual values in this row of data.

6. The third row of a stem-and-leaf chart appears as follows: 21 | 0 1 3 5 7 9. Assume whole number values.
a. What is the "possible range" of the values in this row?
b. How many data values are in this row?
c. List the actual values in this row of data.

7. The following stem-and-leaf chart shows the number of units produced per day in a factory.

Stem	Leaf
3	8
4	
5	6
6	0133559
7	0236778
8	59
9	00156
10	36

a. How many days were studied?
b. How many observations are in the first class?

c. What are the minimum value and the maximum value?
d. List the actual values in the fourth row.
e. List the actual values in the second row.
f. How many values are less than 70?
g. How many values are 80 or more?
h. What is the median?
i. How many values are between 60 and 89, inclusive?

8. The following stem-and-leaf chart reports the number of prescriptions filled per day at the pharmacy on the corner of Fourth and Main Streets.

Stem	Leaf
12	689
13	123
14	6889
15	589
16	35
17	24568
18	268
19	13456
20	034679
21	2239
22	789
23	00179
24	8
25	13
26	
27	0

a. How many days were studied?
b. How many observations are in the last class?
c. What are the maximum and the minimum values in the entire set of data?
d. List the actual values in the fourth row.
e. List the actual values in the next to the last row.
f. On how many days were less than 160 prescriptions filled?
g. On how many days were 220 or more prescriptions filled?
h. What is the middle value?
i. How many days did the number of filled prescriptions range between 170 and 210?

9. FILE A survey of the number of phone calls made by a sample of 16 Verizon subscribers last week revealed the following information. Develop a stem-and-leaf chart. How many calls did a typical subscriber make? What were the maximum and the minimum number of calls made?

52	43	30	38	30	42	12	46	39
37	34	46	32	18	41	5		

10. FILE Aloha Banking Co. is studying ATM use in suburban Honolulu. Yesterday, for a sample of 30 ATM's, the bank counted the number of times each machine was used. The data is presented in the table. Develop a stem-and-leaf chart to summarize the data. What were the typical, minimum, and maximum number of times each ATM was used?

83	64	84	76	84	54	75	59	70	61
63	80	84	73	68	52	65	90	52	77
95	36	78	61	59	84	95	47	87	60

LO4-3
Identify and compute measures of position.

MEASURES OF POSITION

The standard deviation is the most widely used measure of dispersion. However, there are other ways of describing the variation or spread in a set of data. One method is to determine the *location* of values that divide a set of observations into equal parts. These measures include **quartiles, deciles,** and **percentiles.**

Quartiles divide a set of observations into four equal parts. To explain further, think of any set of values arranged from the minimum to the maximum. In Chapter 3, we called the middle value of a set of data arranged from the minimum to the maximum the median. That is, 50% of the observations are larger than the median and 50% are smaller. The median is a measure of location because it pinpoints the center of the data. In a similar fashion, **quartiles** divide a set of observations into four equal parts. The first quartile, usually labeled Q_1, is the value below which 25% of the observations occur, and the third quartile, usually labeled Q_3, is the value below which 75% of the observations occur.

Similarly, **deciles** divide a set of observations into 10 equal parts and **percentiles** into 100 equal parts. So if you found that your GPA was in the 8th decile at your university, you could conclude that 80% of the students had a GPA lower than yours and 20% had a higher GPA. If your GPA was in the 92nd percentile, then 92% of students had a GPA less than your GPA and only 8% of students had a GPA greater than your GPA. Percentile scores are frequently used to report results on such national standardized tests as the SAT, ACT, GMAT (used to judge entry into many master of business administration programs), and LSAT (used to judge entry into law school).

Quartiles, Deciles, and Percentiles

To formalize the computational procedure, let L_p refer to the location of a desired percentile. So if we want to find the 92nd percentile we would use L_{92}, and if we wanted the median, the 50th percentile, then L_{50}. For a number of observations, n, the location of the *Pth* percentile, can be found using the formula:

LOCATION OF A PERCENTILE

$$L_p = (n + 1)\frac{P}{100} \qquad \textbf{[4–1]}$$

An example will help to explain further.

EXAMPLE

Morgan Stanley is an investment company with offices located throughout the United States. Listed below are the commissions earned last month by a sample of 15 brokers at the Morgan Stanley office in Oakland, California.

$2,038	$1,758	$1,721	$1,637	$2,097	$2,047	$2,205	$1,787	$2,287
1,940	2,311	2,054	2,406	1,471	1,460			

Locate the median, the first quartile, and the third quartile for the commissions earned.

SOLUTION

The first step is to sort the data from the smallest commission to the largest.

$1,460	$1,471	$1,637	$1,721	$1,758	$1,787	$1,940	$2,038
2,047	2,054	2,097	2,205	2,287	2,311	2,406	

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The median value is the observation in the center and is the same as the 50th percentile, so P equals 50. So the median or L_{50} is located at $(n + 1)(50/100)$, where n is the number of observations. In this case, that is position number 8, found by $(15 + 1)(50/100)$. The eighth-largest commission is $2,038. So we conclude this is the median and that half the brokers earned commissions more than $2,038 and half earned less than $2,038. The result using formula (4–1) to find the median is the same as the method presented in Chapter 3.

Recall the definition of a quartile. Quartiles divide a set of observations into four equal parts. Hence 25% of the observations will be less than the first quartile. Seventy-five percent of the observations will be less than the third quartile. To locate the first quartile, we use formula (4–1), where $n = 15$ and $P = 25$:

$$L_{25} = (n + 1)\frac{P}{100} = (15 + 1)\frac{25}{100} = 4$$

and to locate the third quartile, $n = 15$ and $P = 75$:

$$L_{75} = (n + 1)\frac{P}{100} = (15 + 1)\frac{75}{100} = 12$$

Therefore, the first and third quartile values are located at positions 4 and 12, respectively. The fourth value in the ordered array is $1,721 and the twelfth is $2,205. These are the first and third quartiles.

In the above example, the location formula yielded a whole number. That is, we wanted to find the first quartile and there were 15 observations, so the location formula indicated we should find the fourth ordered value. What if there were 20 observations in the sample, that is $n = 20$, and we wanted to locate the first quartile? From the location formula (4–1):

$$L_{25} = (n + 1)\frac{P}{100} = (20 + 1)\frac{25}{100} = 5.25$$

We would locate the fifth value in the ordered array and then move .25 of the distance between the fifth and sixth values and report that as the first quartile. Like the median, the quartile does not need to be one of the actual values in the data set.

To explain further, suppose a data set contained the six values 91, 75, 61, 101, 43, and 104. We want to locate the first quartile. We order the values from the minimum to the maximum: 43, 61, 75, 91, 101, and 104. The first quartile is located at

$$L_{25} = (n + 1)\frac{P}{100} = (6 + 1)\frac{25}{100} = 1.75$$

The position formula tells us that the first quartile is located between the first and the second values and it is .75 of the distance between the first and the second values. The first value is 43 and the second is 61. So the distance between these two values is 18. To locate the first quartile, we need to move .75 of the distance between the first and second values, so $.75(18) = 13.5$. To complete the procedure, we add 13.5 to the first value, 43, and report that the first quartile is 56.5.

We can extend the idea to include both deciles and percentiles. To locate the 23rd percentile in a sample of 80 observations, we would look for the 18.63 position.

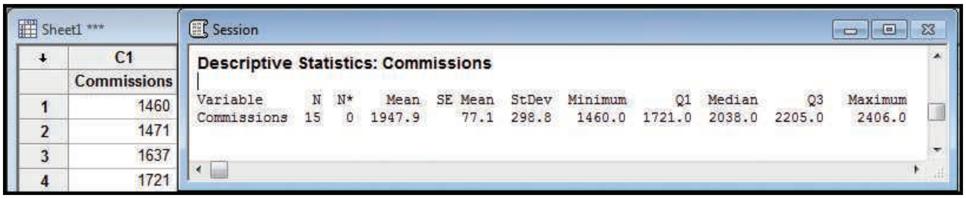
$$L_{23} = (n + 1)\frac{P}{100} = (80 + 1)\frac{23}{100} = 18.63$$

STATISTICS IN ACTION

John W. Tukey (1915–2000) received a PhD in mathematics from Princeton in 1939. However, when he joined the Fire Control Research Office during World War II, his interest in abstract mathematics shifted to applied statistics. He developed effective numerical and graphical methods for studying patterns in data. Among the graphics he developed are the stem-and-leaf diagram and the box-and-whisker plot or box plot. From 1960 to 1980, Tukey headed the statistical division of NBC's election night vote projection team. He became renowned in 1960 for preventing an early call of victory for Richard Nixon in the presidential election won by John F. Kennedy.

To find the value corresponding to the 23rd percentile, we would locate the 18th value and the 19th value and determine the distance between the two values. Next, we would multiply this difference by 0.63 and add the result to the smaller value. The result would be the 23rd percentile.

Statistical software is very helpful when describing and summarizing data. Excel, Minitab, and MegaStat, a statistical analysis Excel add-in, all provide summary statistics that include quartiles. For example, the Minitab summary of the Morgan Stanley commission data, shown below, includes the first and third quartiles, and other statistics. Based on the reported quartiles, 25% of the commissions earned were less than \$1,721 and 75% were less than \$2,205. These are the same values we calculated using formula (4–1).



Morgan Stanley Commissions		
1460	Equation 4-1	
2047	Quartile 1	1721
1471	Quartile 3	2205
2054		
1637		
2097	Alternate Method	
1721	Quartile 1	1739.5
2205	Quartile 3	2151
1758		
2287		
1787		
2311		
1940		
2406		
2038		

There are ways other than formula (4–1) to locate quartile values. For example, another method uses $0.25n + 0.75$ to locate the position of the first quartile and $0.75n + 0.25$ to locate the position of the third quartile. We will call this the *Excel Method*. In the Morgan Stanley data, this method would place the first quartile at position 4.5 (.25 × 15 + .75) and the third quartile at position 11.5 (.75 × 15 + .25). The first quartile would be interpolated as 0.5, or one-half the difference between the fourth- and the fifth-ranked values. Based on this method, the first quartile is \$1739.5, found by (\$1,721 + 0.5[\$1,758 – \$1,721]). The third quartile, at position 11.5, would be \$2,151, or one-half the distance between the eleventh- and the twelfth-ranked values, found by (\$2,097 + 0.5[\$2,205 – \$2,097]). Excel, as shown in the Morgan Stanley and Applewood examples, can compute quartiles using either of the two methods. **Please note the text uses formula (4–1) to calculate quartiles.**

Is the difference between the two methods important? No. Usually it is just a nuisance. In general, both methods calculate values that will support the statement that approximately 25% of the values are less than the value of the first quartile, and approximately 75% of the data values are less than the value of the third quartile. When the sample is large, the difference in the results from the two methods is small. For example, in the Applewood Auto Group data there are 180 vehicles. The quartiles computed using both methods are shown to the left. Based on the variable profit, 45 of the 180 values (25%) are less than both values of the first quartile, and 135 of the 180 values (75%) are less than both values of the third quartile.

Applewood				
Age	Profit			
21	\$1,387			
23	\$1,754			
24	\$1,817		Equation 4-1	
25	\$1,040		Quartile 1	1415.5
26	\$1,273		Quartile 3	2275.5
27	\$1,529			
27	\$3,082		Alternate Method	
28	\$1,951		Quartile 1	1422.5
28	\$2,692		Quartile 3	2268.5
29	\$1,342			

When using Excel, be careful to understand the method used to calculate quartiles. Excel 2013 and Excel 2016 offer both methods. The Excel function, ***Quartile.exc,*** will result in the same answer as Equation 4–1. The Excel function, *Quartile.inc*, will result in the Excel Method answers.

SELF-REVIEW 4–2

The Quality Control department of Plainsville Peanut Company is responsible for checking the weight of the 8-ounce jar of peanut butter. The weights of a sample of nine jars produced last hour are:

7.69	7.72	7.8	7.86	7.90	7.94	7.97	8.06	8.09

(a) What is the median weight?
(b) Determine the weights corresponding to the first and third quartiles.

EXERCISES

11. FILE Determine the median and the first and third quartiles in the following data.

46	47	49	49	51	53	54	54	55	55	59

12. FILE Determine the median and the first and third quartiles in the following data.

5.24	6.02	6.67	7.30	7.59	7.99	8.03	8.35	8.81	9.45
9.61	10.37	10.39	11.86	12.22	12.71	13.07	13.59	13.89	15.42

13. FILE The Thomas Supply Company Inc. is a distributor of gas-powered generators. As with any business, the length of time customers take to pay their invoices is important. Listed below, arranged from smallest to largest, is the time, in days, for a sample of The Thomas Supply Company Inc. invoices.

13	13	13	20	26	27	31	34	34	34	35	35	36	37	38
41	41	41	45	47	47	47	50	51	53	54	56	62	67	82

a. Determine the first and third quartiles.
b. Determine the second decile and the eighth decile.
c. Determine the 67th percentile.

14. FILE Kevin Horn is the national sales manager for National Textbooks Inc. He has a sales staff of 40 who visit college professors all over the United States. Each Saturday morning he requires his sales staff to send him a report. This report includes, among other things, the number of professors visited during the previous week. Listed below, ordered from smallest to largest, are the number of visits last week.

38	40	41	45	48	48	50	50	51	51	52	52	53	54	55	55	55	56	56	57
59	59	59	62	62	62	63	64	65	66	66	67	67	69	69	71	77	78	79	79

a. Determine the median number of calls.
b. Determine the first and third quartiles.
c. Determine the first decile and the ninth decile.
d. Determine the 33rd percentile.

LO4-4
Construct and analyze a box plot.

BOX PLOTS

A **box plot** is a graphical display, based on quartiles, that helps us picture a set of data. To construct a box plot, we need only five statistics: the minimum value, Q_1 (the first quartile), the median, Q_3 (the third quartile), and the maximum value. An example will help to explain.

EXAMPLE

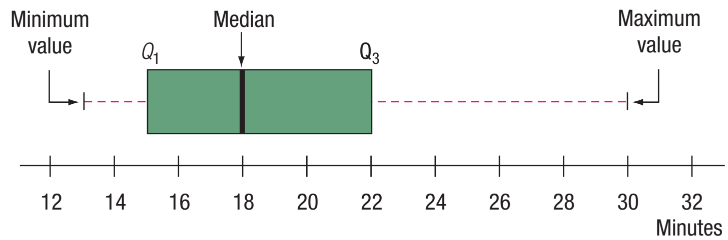
Alexander's Pizza offers free delivery of its pizza within 15 miles. Alex, the owner, wants some information on the time it takes for delivery. How long does a typical delivery take? Within what range of times will most deliveries be completed? For a sample of 20 deliveries, he determined the following information:

$$\text{Minimum value} = 13 \text{ minutes}$$
$$Q_1 = 15 \text{ minutes}$$
$$\text{Median} = 18 \text{ minutes}$$
$$Q_3 = 22 \text{ minutes}$$
$$\text{Maximum value} = 30 \text{ minutes}$$

Develop a box plot for the delivery times. What conclusions can you make about the delivery times?

SOLUTION

The first step in drawing a box plot is to create an appropriate scale along the horizontal axis. Next, we draw a box that starts at Q_1 (15 minutes) and ends at Q_3 (22 minutes). Inside the box we place a vertical line to represent the median (18 minutes). Finally, we extend horizontal lines from the box out to the minimum value (13 minutes) and the maximum value (30 minutes). These horizontal lines outside of the box are sometimes called "whiskers" because they look a bit like a cat's whiskers.



The box plot also shows the interquartile range of delivery times between Q1 and Q3. The **interquartile range** is 7 minutes and indicates that 50% of the deliveries are between 15 and 22 minutes.

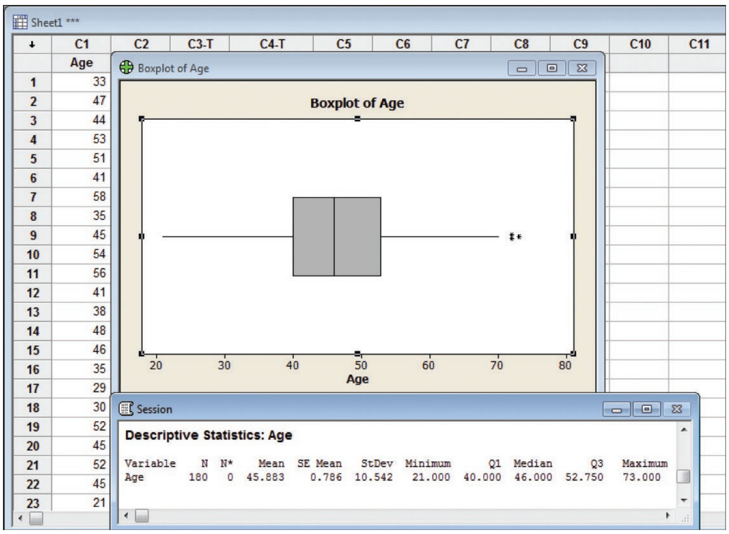
The box plot also reveals that the distribution of delivery times is positively skewed. In Chapter 3, we defined skewness as the lack of symmetry in a set of data. How do we know this distribution is positively skewed? In this case, there are actually two pieces of information that suggest this. First, the dashed line to the right of the box from 22 minutes (Q_3) to the maximum time of 30 minutes is longer than the dashed line from the left of 15 minutes (Q_1) to the minimum value of 13 minutes. To put it another way, the 25% of the data larger than the third quartile is more spread out than the 25% less than the first quartile. A second indication of positive skewness is that the median is not in the center of the box. The distance from the first quartile to the median is smaller than the distance from the median to the third quartile. We know that the number of delivery times between 15 minutes and 18 minutes is the same as the number of delivery times between 18 minutes and 22 minutes.

EXAMPLE

Refer to the Applewood Auto Group data. Develop a box plot for the variable age of the buyer. What can we conclude about the distribution of the age of the buyer?

SOLUTION

Minitab was used to develop the following chart and summary statistics.



The median age of the purchaser is 46 years, 25% of the purchasers are less than 40 years of age, and 25% are more than 52.75 years of age. Based on the summary information and the box plot, we conclude:

- Fifty percent of the purchasers are between the ages of 40 and 52.75 years.
- The distribution of ages is fairly symmetric. There are two reasons for this conclusion. The length of the whisker above 52.75 years (Q_3) is about the same length as the whisker below 40 years (Q_1). Also, the area in the box between 40 years and the median of 46 years is about the same as the area between the median and 52.75.

There are three asterisks (*) above 70 years. What do they indicate? In a box plot, an asterisk identifies an **outlier.** An outlier is a value that is inconsistent with the rest of the data. It is defined as a value that is more than 1.5 times the interquartile range smaller than Q_1 or larger than Q_3. In this example, an outlier would be a value larger than 71.875 years, found by:

$$\textit{Outlier} > Q_3 + 1.5(Q_3 - Q_1) = 52.75 + 1.5(52.75 - 40) = 71.875$$

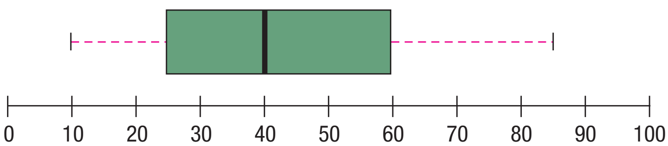
An outlier would also be a value less than 20.875 years.

$$\textit{Outlier} < Q_1 - 1.5(Q_3 - Q_1) = 40 - 1.5(52.75 - 40) = 20.875$$

From the box plot, we conclude there are three purchasers 72 years of age or older and none less than 21 years of age. Technical note: In some cases, a single asterisk may represent more than one observation because of the limitations of the software and space available. It is a good idea to check the actual data. In this instance, there are three purchasers 72 years old or older; two are 72 and one is 73.

SELF-REVIEW 4–3

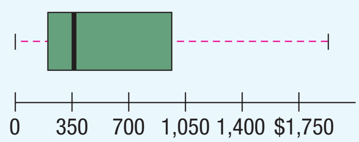
The following box plot shows the assets in millions of dollars for credit unions in Seattle, Washington.



What are the smallest and largest values, the first and third quartiles, and the median? Would you agree that the distribution is symmetrical? Are there any outliers?

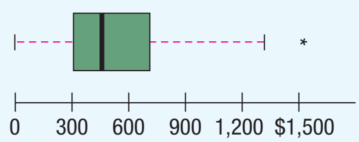
EXERCISES

15. The box plot below shows the amount spent for books and supplies per year by students at four-year public colleges.



a. Estimate the median amount spent.
b. Estimate the first and third quartiles for the amount spent.
c. Estimate the interquartile range for the amount spent.
d. Beyond what point is a value considered an outlier?
e. Identify any outliers and estimate their value.
f. Is the distribution symmetrical or positively or negatively skewed?

16. The box plot shows the undergraduate in-state tuition per credit hour at four-year public colleges.



a. Estimate the median.
b. Estimate the first and third quartiles.
c. Determine the interquartile range.
d. Beyond what point is a value considered an outlier?
e. Identify any outliers and estimate their value.
f. Is the distribution symmetrical or positively or negatively skewed?

17. In a study of the gasoline mileage of model year 2016 automobiles, the mean miles per gallon was 27.5 and the median was 26.8. The smallest value in the study was 12.70 miles per gallon, and the largest was 50.20. The first and third quartiles were 17.95 and 35.45 miles per gallon, respectively. Develop a box plot and comment on the distribution. Is it a symmetric distribution?

18. **FILE** A sample of 28 time shares in the Orlando, Florida, area revealed the following daily charges for a one-bedroom suite. For convenience, the data are ordered from smallest to largest. Construct a box plot to represent the data. Comment on the distribution. Be sure to identify the first and third quartiles and the median.

$116	$121	$157	$192	$207	$209	$209
229	232	236	236	239	243	246
260	264	276	281	283	289	296
307	309	312	317	324	341	353

LO4-5
Compute and interpret the coefficient of skewness.

SKEWNESS

In Chapter 3, we described measures of central location for a distribution of data by reporting the mean, median, and mode. We also described measures that show the amount of spread or variation in a distribution, such as the range and the standard deviation.

Another characteristic of a distribution is the shape. There are four shapes commonly observed: symmetric, positively skewed, negatively skewed, and bimodal. In a **symmetric** distribution the mean and median are equal and the data values are evenly spread around these values. The shape of the distribution below the mean and median is a mirror image of distribution above the mean and median. A distribution of values is **skewed to the right** or **positively skewed** if there is a single peak, but the values extend much farther to the right of the peak than to the left of the peak. In this case, the mean is larger than the median. In a **negatively skewed** distribution there is a single peak, but the observations extend farther to the left, in the negative direction, than to the right. In a negatively skewed distribution, the mean is smaller than the median. Positively skewed distributions are more common. Salaries often follow this pattern. Think of the salaries of those employed in a small company of about 100 people. The president and a few top executives would have very large salaries relative to the other workers and hence the distribution of salaries would exhibit positive skewness. A **bimodal distribution** will have two or more peaks. This is often the case when the values are from two or more populations. This information is summarized in Chart 4–1.

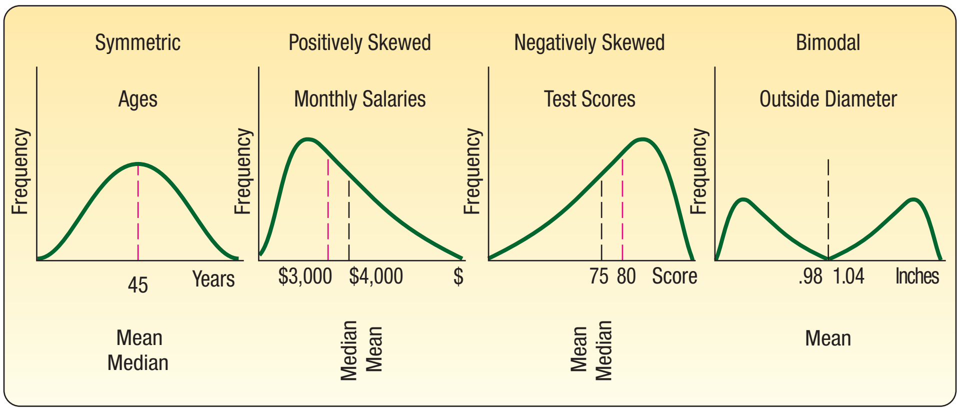


CHART 4–1 Shapes of Frequency Polygons

There are several formulas in the statistical literature used to calculate skewness. The simplest, developed by Professor Karl Pearson (1857–1936), is based on the difference between the mean and the median.

STATISTICS IN ACTION

The late Stephen Jay Gould (1941–2002) was a professor of zoology and professor of geology at Harvard University. In 1982, he was diagnosed with cancer and had an expected survival time of 8 months. However, never to be discouraged, his research showed that the distribution of survival time is dramatically skewed to the right and showed that not only do 50% of similar cancer patients survive more than 8 months, but that the survival time could be years rather than months! In fact, Dr. Gould lived another 20 years. Based on his experience, he wrote a widely published essay titled "The Median Is Not the Message."

PEARSON'S COEFFICIENT OF SKEWNESS

$$sk = \frac{3(\bar{x} - \text{Median})}{s} \qquad \textbf{[4–2]}$$

Using this relationship, the coefficient of skewness can range from −3 up to 3. A value near −3, such as −2.57, indicates considerable negative skewness. A value such as 1.63 indicates moderate positive skewness. A value of 0, which will occur when the mean and median are equal, indicates the distribution is symmetrical and there is no skewness present.

In this text, we present output from Minitab and Excel. Both of these software packages compute a value for the coefficient of skewness based on the cubed deviations from the mean. The formula is:

SOFTWARE COEFFICIENT OF SKEWNESS

$$sk = \frac{n}{(n-1)(n-2)}\left[\sum\left(\frac{x-\bar{x}}{s}\right)^3\right] \qquad \textbf{[4–3]}$$

Formula (4–3) offers an insight into skewness. The right-hand side of the formula is the difference between each value and the mean, divided by the standard deviation. That is the portion $(x - \bar{x})/s$ of the formula. This idea is called **standardizing.** We will discuss the idea of standardizing a value in more detail in Chapter 7 when we describe the normal probability distribution. At this point, observe that the result is to report the difference between each value and the mean in units of the standard deviation. If this difference is positive, the particular value is larger than the mean; if the value is negative, the standardized quantity is smaller than the mean. When we cube these values, we retain the information on the direction of the difference. Recall that in the formula for the standard deviation [see formula (3–10)] we squared the difference between each value and the mean, so that the result was all nonnegative values.

If the set of data values under consideration is symmetric, when we cube the standardized values and sum over all the values, the result would be near zero. If there are several large values, clearly separate from the others, the sum of the cubed differences would be a large positive value. If there are several small values clearly separate from the others, the sum of the cubed differences will be negative.

An example will illustrate the idea of skewness.

EXAMPLE

Following are the earnings per share for a sample of 15 software companies for the year 2016. The earnings per share are arranged from smallest to largest.

$0.09	$0.13	$0.41	$0.51	$ 1.12	$ 1.20	$ 1.49	$3.18
3.50	6.36	7.83	8.92	10.13	12.99	16.40	

Compute the mean, median, and standard deviation. Find the coefficient of skewness using Pearson's estimate and the software methods. What is your conclusion regarding the shape of the distribution?

SOLUTION

These are sample data, so we use formula (3–2) to determine the mean

$$\bar{x} = \frac{\Sigma x}{n} = \frac{\$74.26}{15} = \$4.95$$

The median is the middle value in a set of data, arranged from smallest to largest. In this case, there is an odd-number of observations, so the middle value is the median. It is \$3.18.

We use formula (3–10) on page 78 to determine the sample standard deviation.

$$s = \sqrt{\frac{\Sigma(x - \bar{x})^2}{n - 1}} = \sqrt{\frac{(\$0.09 - \$4.95)^2 + \cdots + (\$16.40 - \$4.95)^2}{15 - 1}} = \$5.22$$

Pearson's coefficient of skewness is 1.017, found by

$$sk = \frac{3(\bar{x} - \text{Median})}{s} = \frac{3(\$4.95 - \$3.18)}{\$5.22} = 1.017$$

This indicates there is moderate positive skewness in the earnings per share data.

We obtain a similar, but not exactly the same, value from the software method. The details of the calculations are shown in Table 4–2. To begin, we find the difference between each earnings per share value and the mean and divide this result by the standard deviation. We have referred to this as standardizing. Next, we cube, that is, raise to the third power, the result of the first step. Finally, we sum the cubed values. The details for the first company, that is, the company with an earnings per share of \$0.09, are:

$$\left(\frac{x - \bar{x}}{s}\right)^3 = \left(\frac{0.09 - 4.95}{5.22}\right)^3 = (-0.9310)^3 = -0.8070$$

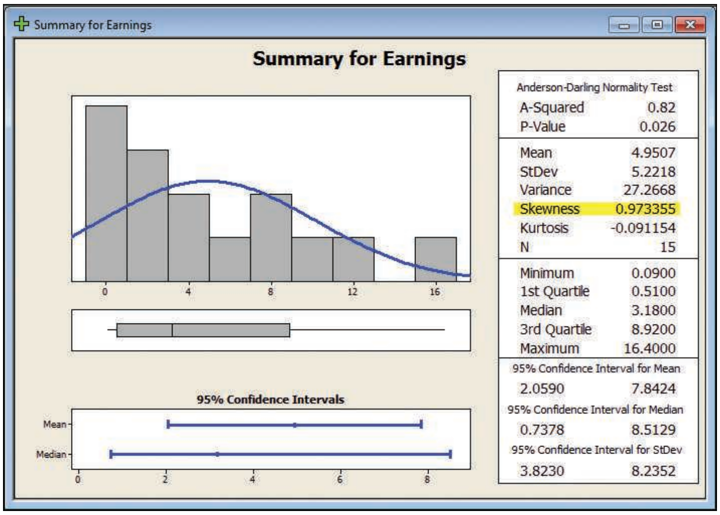
TABLE 4–2 Calculation of the Coefficient of Skewness

Earnings per Share	$\frac{(x - \bar{x})}{s}$	$\left(\frac{x - \bar{x}}{s}\right)^3$
0.09	−0.9310	−0.8070
0.13	−0.9234	−0.7873
0.41	−0.8697	−0.6579
0.51	−0.8506	−0.6154
1.12	−0.7337	−0.3950
1.20	−0.7184	−0.3708
1.49	−0.6628	−0.2912
3.18	−0.3391	−0.0390
3.50	−0.2778	−0.0214
6.36	0.2701	0.0197
7.83	0.5517	0.1679
8.92	0.7605	0.4399
10.13	0.9923	0.9772
12.99	1.5402	3.6539
16.40	2.1935	10.5537
		11.8274

When we sum the 15 cubed values, the result is 11.8274. That is, the term $\Sigma[(x - \bar{x})/s]^3 = 11.8274$. To find the coefficient of skewness, we use formula (4–3), with $n = 15$.

$$sk = \frac{n}{(n - 1)(n - 2)} \sum \left(\frac{x - \bar{x}}{s}\right)^3 = \frac{15}{(15 - 1)(15 - 2)}(11.8274) = 0.975$$

We conclude that the earnings per share values are somewhat positively skewed. The following Minitab summary reports the descriptive measures, such as the mean, median, and standard deviation of the earnings per share data. Also included are the coefficient of skewness and a histogram with a bell-shaped curve superimposed.



SELF-REVIEW 4–4

A sample of five data entry clerks employed in the Horry County Tax Office revised the following number of tax records last hour: 73, 98, 60, 92, and 84.

(a) Find the mean, median, and the standard deviation.
(b) Compute the coefficient of skewness using Pearson's method.
(c) Calculate the coefficient of skewness using the software method.
(d) What is your conclusion regarding the skewness of the data?

EXERCISES

For Exercises 19–22:

a. Determine the mean, median, and the standard deviation.
b. Determine the coefficient of skewness using Pearson's method.
c. Determine the coefficient of skewness using the software method.

19. **FILE** The following values are the starting salaries, in $000, for a sample of five accounting graduates who accepted positions in public accounting last year.

36.0	26.0	33.0	28.0	31.0

20. **FILE** Listed below are the salaries, in $000, for a sample of 15 chief financial officers in the electronics industry.

$516.0	$548.0	$566.0	$534.0	$586.0	$529.0
546.0	523.0	538.0	523.0	551.0	552.0
486.0	558.0	574.0			

21. FILE Listed below are the commissions earned ($000) last year by the 15 sales representatives at Furniture Patch Inc.

$ 3.9	$ 5.7	$ 7.3	$10.6	$13.0	$13.6	$15.1	$15.8	$17.1
17.4	17.6	22.3	38.6	43.2	87.7			

22. FILE Listed below are the salaries for the 2016 New York Yankees Major League Baseball team.

Player	Salary	Player	Salary
CC Sabathia	$25,000,000	Dustin Ackley	$3,200,000
Mark Teixeira	23,125,000	Martin Prado	3,000,000
Masahiro Tanaka	22,000,000	Didi Gregorius	2,425,000
Jacoby Ellsbury	21,142,857	Aaron Hicks	574,000
Alex Rodriguez	21,000,000	Austin Romine	556,000
Brian McCann	17,000,000	Chasen Shreve	533,400
Carlos Beltran	15,000,000	Greg Bird	525,300
Brett Gardner	13,500,000	Luis Severino	521,300
Chase Headley	13,000,000	Bryan Mitchell	516,650
Aroldis Chapman	11,325,000	Kirby Yates	511,900
Andrew Miller	9,000,000	Mason Williams	509,700
Starlin Castro	7,857,143	Ronald Torreyes	508,600
Nathan Eovaldi	5,600,000	John Barbato	507,500
Michael Pineda	4,300,000	Dellin Betances	507,500
Ivan Nova	4,100,000	Luis Cessa	507,500

LO4-6
Create and interpret a scatter diagram.

DESCRIBING THE RELATIONSHIP BETWEEN TWO VARIABLES

In Chapter 2 and the first section of this chapter, we presented graphical techniques to summarize the distribution of a single variable. We used a histogram in Chapter 2 to summarize the profit on vehicles sold by the Applewood Auto Group. Earlier in this chapter, we used dot plots and stem-and-leaf displays to visually summarize a set of data. Because we are studying a single variable, we refer to this as **univariate** data.

© Steve Mason/Getty Images RF

There are situations where we wish to study and visually portray the relationship between two variables. When we study the relationship between two variables, we refer to the data as **bivariate.** Data analysts frequently wish to understand the relationship between two variables. Here are some examples:

- Tybo and Associates is a law firm that advertises extensively on local TV. The partners are considering increasing their advertising budget. Before doing so, they would like to know the relationship between the amount spent per month on advertising and the total amount of billings for that month. To put it another way, will increasing the amount spent on advertising result in an increase in billings?

- Coastal Realty is studying the selling prices of homes. What variables seem to be related to the selling price of homes? For example, do larger homes sell for more than smaller ones? Probably. So Coastal might study the relationship between the area in square feet and the selling price.
- Dr. Stephen Givens is an expert in human development. He is studying the relationship between the height of fathers and the height of their sons. That is, do tall fathers tend to have tall children? Would you expect LeBron James, the 6′8″, 250 pound professional basketball player, to have relatively tall sons?

One graphical technique we use to show the relationship between variables is called a **scatter diagram.**

To draw a scatter diagram, we need two variables. We scale one variable along the horizontal axis (*X*-axis) of a graph and the other variable along the vertical axis (*Y*-axis). Usually one variable depends to some degree on the other. In the third example above, the height of the son *depends* on the height of the father. So we scale the height of the father on the horizontal axis and that of the son on the vertical axis.

We can use statistical software, such as Excel, to perform the plotting function for us. *Caution:* You should always be careful of the scale. By changing the scale of either the vertical or the horizontal axis, you can affect the apparent visual strength of the relationship.

Following are three scatter diagrams (Chart 4–2). The one on the left shows a rather strong positive relationship between the age in years and the maintenance cost last year for a sample of 10 buses owned by the city of Cleveland, Ohio. Note that as the age of the bus increases, the yearly maintenance cost also increases. The example in the center, for a sample of 20 vehicles, shows a rather strong indirect relationship between the odometer reading and the auction price. That is, as the number of miles driven increases, the auction price decreases. The example on the right depicts the relationship between the height and yearly salary for a sample of 15 shift supervisors. This graph indicates there is little relationship between their height and yearly salary.

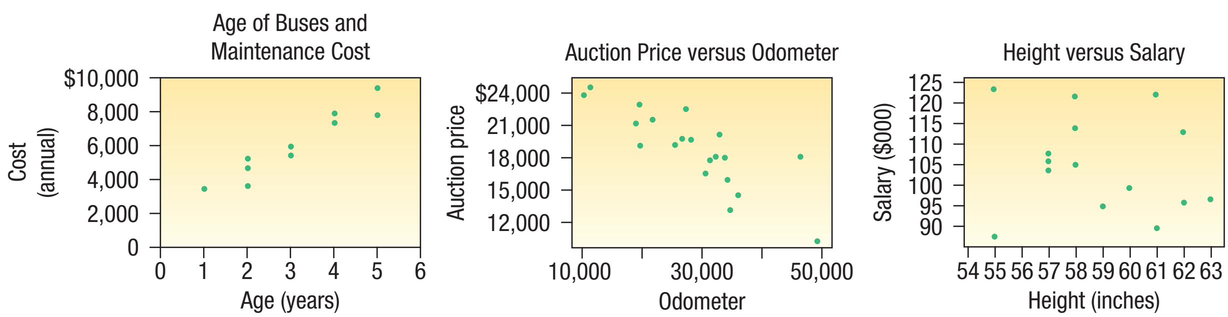


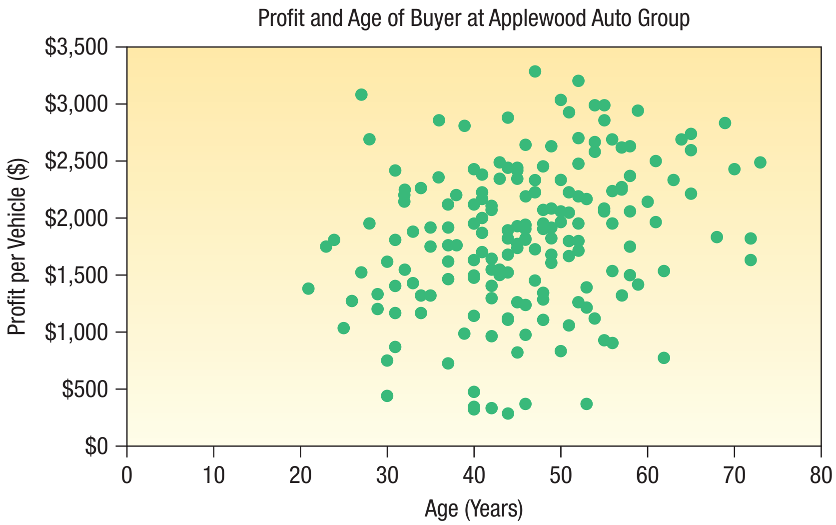
CHART 4–2 Three Examples of Scatter Diagrams.

EXAMPLE

In the introduction to Chapter 2, we presented data from the Applewood Auto Group. We gathered information concerning several variables, including the profit earned from the sale of 180 vehicles sold last month. In addition to the amount of profit on each sale, one of the other variables is the age of the purchaser. Is there a relationship between the profit earned on a vehicle sale and the age of the purchaser? Would it be reasonable to conclude that more profit is made on vehicles purchased by older buyers?

SOLUTION

We can investigate the relationship between vehicle profit and the age of the buyer with a scatter diagram. We scale age on the horizontal, or *X*-axis, and the profit on the vertical, or *Y*-axis. We assume profit depends on the age of the purchaser. As people age, they earn more income and purchase more expensive cars which, in turn, produce higher profits. We use Excel to develop the scatter diagram. The Excel commands are in Appendix C.



The scatter diagram shows a rather weak positive relationship between the two variables. It does not appear there is much relationship between the vehicle profit and the age of the buyer. In Chapter 13, we will study the relationship between variables more extensively, even calculating several numerical measures to express the relationship between variables.

In the preceding example, there is a weak positive, or direct, relationship between the variables. There are, however, many instances where there is a relationship between the variables, but that relationship is inverse or negative. For example:

- The value of a vehicle and the number of miles driven. As the number of miles increases, the value of the vehicle decreases.
- The premium for auto insurance and the age of the driver. Auto rates tend to be the highest for younger drivers and less for older drivers.
- For many law enforcement personnel, as the number of years on the job increases, the number of traffic citations decreases. This may be because personnel become more liberal in their interpretations or they may be in supervisor positions and not in a position to issue as many citations. But in any event, as age increases, the number of citations decreases.

LO4-7
Develop and explain a contingency table.

CONTINGENCY TABLES

A scatter diagram requires that both of the variables be at least interval scale. In the Applewood Auto Group example, both age and vehicle profit are ratio scale variables. Height is also ratio scale as used in the discussion of the relationship between the height of fathers and the height of their sons. What if we wish to study the relationship between two variables when one or both are nominal or ordinal scale? In this case, we tally the results in a **contingency table.**

CONTINGENCY TABLE A table used to classify observations according to two identifiable characteristics.

A contingency table is a cross-tabulation that simultaneously summarizes two variables of interest. For example:

- Students at a university are classified by gender and class (freshman, sophomore, junior, or senior).
- A product is classified as acceptable or unacceptable and by the shift (day, afternoon, or night) on which it is manufactured.
- A voter in a school bond referendum is classified as to party affiliation (Democrat, Republican, other) and the number of children that voter has attending school in the district (0, 1, 2, etc.).

EXAMPLE

There are four dealerships in the Applewood Auto Group. Suppose we want to compare the profit earned on each vehicle sold by the particular dealership. To put it another way, is there a relationship between the amount of profit earned and the dealership?

SOLUTION

In a contingency table, both variables only need to be nominal or ordinal. In this example, the variable dealership is a nominal variable and the variable profit is a ratio variable. To convert profit to an ordinal variable, we classify the variable profit into two categories, those cases where the profit earned is more than the median and those cases where it is less. On page 64, we calculated the median profit for all sales last month at Applewood Auto Group to be $1,882.50.

Contingency Table Showing the Relationship between Profit and Dealership

Above/Below Median Profit	Kane	Olean	Sheffield	Tionesta	Total
Above	25	20	19	26	90
Below	27	20	26	17	90
Total	52	40	45	43	180

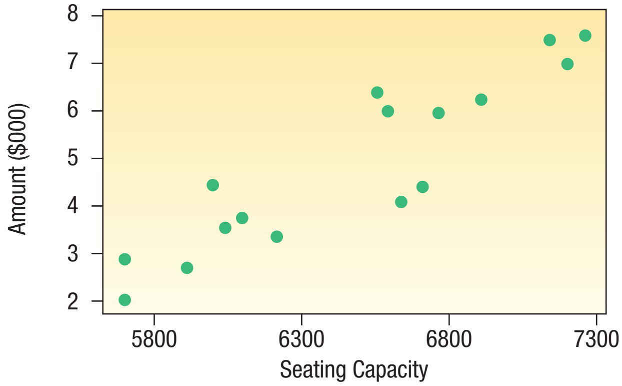
By organizing the information into a contingency table, we can compare the profit at the four dealerships. We observe the following:

- From the Total column on the right, 90 of the 180 cars sold had a profit above the median and half below. From the definition of the median, this is expected.
- For the Kane dealership, 25 out of the 52, or 48%, of the cars sold were sold for a profit more than the median.
- The percentage of profits above the median for the other dealerships are 50% for Olean, 42% for Sheffield, and 60% for Tionesta.

We will return to the study of contingency tables in Chapter 5 during the study of probability and in Chapter 15 during the study of nonparametric methods of analysis.

SELF-REVIEW 4–5

The rock group Blue String Beans is touring the United States. The following chart shows the relationship between concert seating capacity and revenue in $000 for a sample of concerts.



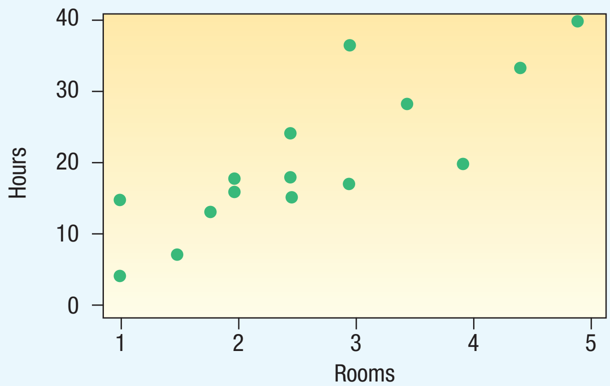
(a) What is the diagram called?
(b) How many concerts were studied?
(c) Estimate the revenue for the concert with the largest seating capacity.
(d) How would you characterize the relationship between revenue and seating capacity? Is it strong or weak, direct or inverse?

EXERCISES

23. FILE Develop a scatter diagram for the following sample data. How would you describe the relationship between the values?

x-Value	*y*-Value	*x*-Value	*y*-Value
10	6	11	6
8	2	10	5
9	6	7	2
11	5	7	3
13	7	11	7

24. Silver Springs Moving and Storage Inc. is studying the relationship between the number of rooms in a move and the number of labor hours required for the move. As part of the analysis, the CFO of Silver Springs developed the following scatter diagram.



a. How many moves are in the sample?
b. Does it appear that more labor hours are required as the number of rooms increases, or do labor hours decrease as the number of rooms increases?

25. The Director of Planning for Devine Dining Inc. wishes to study the relationship between the gender of a guest and whether the guest orders dessert. To investigate the relationship, the manager collected the following information on 200 recent customers.

	Gender		
Dessert Ordered	**Male**	**Female**	**Total**
Yes	32	15	47
No	68	85	153
Total	100	100	200

a. What is the level of measurement of the two variables?
b. What is the above table called?
c. Does the evidence in the table suggest men are more likely to order dessert than women? Explain why.

26. Ski Resorts of Vermont Inc. is considering a merger with Gulf Shores Beach Resorts Inc. of Alabama. The board of directors surveyed 50 stockholders concerning their position on the merger. The results are reported below.

	Opinion			
Number of Shares Held	**Favor**	**Oppose**	**Undecided**	**Total**
Under 200	8	6	2	16
200 up to 1,000	6	8	1	15
Over 1,000	6	12	1	19
Total	20	26	4	50

a. What level of measurement is used in this table?
b. What is this table called?
c. What group seems most strongly opposed to the merger?

CHAPTER SUMMARY

I. A dot plot shows the range of values on the horizontal axis and the number of observations for each value on the vertical axis.
  A. Dot plots report the details of each observation.
  B. They are useful for comparing two or more data sets.
II. A stem-and-leaf display is an alternative to a histogram.
  A. The leading digit is the stem and the trailing digit the leaf.
  B. The advantages of a stem-and-leaf display over a histogram include:
    1. The identity of each observation is not lost.
    2. The digits themselves give a picture of the distribution.
    3. The cumulative frequencies are also shown.
III. Measures of location also describe the shape of a set of observations.
  A. Quartiles divide a set of observations into four equal parts.
    1. Twenty-five percent of the observations are less than the first quartile, 50% are less than the second quartile, and 75% are less than the third quartile.
    2. The interquartile range is the difference between the third quartile and the first quartile.
  B. Deciles divide a set of observations into 10 equal parts and percentiles into 100 equal parts.

IV. A box plot is a graphic display of a set of data.
  A. A box is drawn enclosing the regions between the first quartile and the third quartile.
    1. A line is drawn inside the box at the median value.
    2. Dotted line segments are drawn from the third quartile to the largest value to show the highest 25% of the values and from the first quartile to the smallest value to show the lowest 25% of the values.
  B. A box plot is based on five statistics: the maximum and minimum values, the first and third quartiles, and the median.

V. The coefficient of skewness is a measure of the symmetry of a distribution.
  A. There are two formulas for the coefficient of skewness.
    1. The formula developed by Pearson is:

$$sk = \frac{3(\bar{x} - \text{Median})}{s} \qquad \textbf{[4–2]}$$

    2. The coefficient of skewness computed by statistical software is:

$$sk = \frac{n}{(n-1)(n-2)}\left[\Sigma\left(\frac{x-\bar{x}}{s}\right)^3\right] \qquad \textbf{[4–3]}$$

VI. A scatter diagram is a graphic tool to portray the relationship between two variables.
  A. Both variables are measured with interval or ratio scales.
  B. If the scatter of points moves from the lower left to the upper right, the variables under consideration are directly or positively related.
  C. If the scatter of points moves from the upper left to the lower right, the variables are inversely or negatively related.

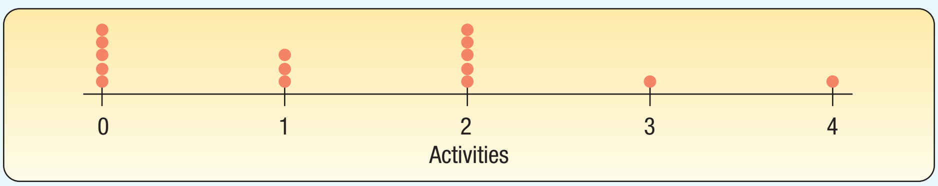
VII. A contingency table is used to classify nominal-scale observations according to two characteristics.

PRONUNCIATION KEY

SYMBOL	MEANING	PRONUNCIATION
L_p	Location of percentile	*L* sub *p*
Q_1	First quartile	*Q* sub 1
Q_3	Third quartile	*Q* sub 3

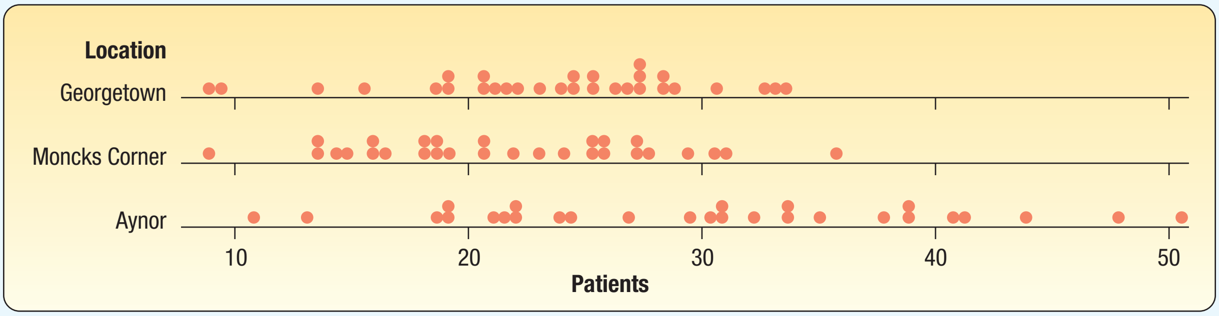
CHAPTER EXERCISES

27. A sample of students attending Southeast Florida University is asked the number of social activities in which they participated last week. The chart below was prepared from the sample data.



  a. What is the name given to this chart?
  b. How many students were in the study?
  c. How many students reported attending no social activities?

28. Doctor's Care is a walk-in clinic, with locations in Georgetown, Moncks Corner, and Aynor, at which patients may receive treatment for minor injuries, colds, and flu, as well as physical examinations. The following charts report the number of patients treated in each of the three locations last month.



Describe the number of patients served at the three locations each day. What are the maximum and minimum numbers of patients served at each of the locations?

29. **FILE** Below is the number of customers who visited Smith's True-Value hardware store in Bellville, Ohio, over the last twenty-three days. Make a stem-and-leaf display of this variable.

46	52	46	40	42	46	40	37	46	40	52	32	37	32	52
40	32	52	40	52	46	46	52							

30. **FILE** The top 25 companies (by market capitalization) operating in the Washington, DC, area along with the year they were founded and the number of employees are given below. Make a stem-and-leaf display of each of these variables and write a short description of your findings.

Company Name	Year Founded	Employees
AES Corp.	1981	30,000
American Capital Ltd.	1986	484
AvalonBay Communities Inc.	1978	1,767
Capital One Financial Corp.	1995	31,800
Constellation Energy Group Inc.	1816	9,736
Coventry Health Care Inc.	1986	10,250
Danaher Corp.	1984	45,000
Dominion Resources Inc.	1909	17,500
Fannie Mae	1938	6,450
Freddie Mac	1970	5,533
Gannett Co.	1906	49,675
General Dynamics Corp.	1952	81,000
Genworth Financial Inc.	2004	7,200
Harman International Industries Inc.	1980	11,246
Host Hotels & Resorts Inc.	1927	229
Legg Mason	1899	3,800
Lockheed Martin Corp.	1995	140,000
Marriott International Inc.	1927	151,000
MedImmune LLC	1988	2,516
NII Holdings Inc.	1996	7,748
Norfolk Southern Corp.	1982	30,594
Pepco Holdings Inc.	1896	5,057
Sallie Mae	1972	11,456
T. Rowe Price Group Inc.	1937	4,605
The Washington Post Co.	1877	17,100

31. **FILE** In recent years, due to low interest rates, many homeowners refinanced their home mortgages. Linda Lahey is a mortgage officer at Down River Federal Savings and Loan. Below is the amount refinanced for 20 loans she processed last week. The data are reported in thousands of dollars and arranged from smallest to largest.

59.2	59.5	61.6	65.5	66.6	72.9	74.8	77.3	79.2
83.7	85.6	85.8	86.6	87.0	87.1	90.2	93.3	98.6
100.2	100.7							

a. Find the median, first quartile, and third quartile.
b. Find the 26th and 83rd percentiles.
c. Draw a box plot of the data.

32. FILE A study is made by the recording industry in the United States of the number of music CDs owned by 25 senior citizens and 30 young adults. The information is reported below.

Seniors

28	35	41	48	52	81	97	98	98	99
118	132	133	140	145	147	153	158	162	174
177	180	180	187	188					

Young Adults

81	107	113	147	147	175	183	192	202	209
233	251	254	266	283	284	284	316	372	401
417	423	490	500	507	518	550	557	590	594

a. Find the median and the first and third quartiles for the number of CDs owned by senior citizens. Develop a box plot for the information.
b. Find the median and the first and third quartiles for the number of CDs owned by young adults. Develop a box plot for the information.
c. Compare the number of CDs owned by the two groups.

33. FILE The corporate headquarters of *Bank.com,* an on-line banking company, is located in downtown Philadelphia. The director of human resources is making a study of the time it takes employees to get to work. The city is planning to offer incentives to each downtown employer if they will encourage their employees to use public transportation. Below is a listing of the time to get to work this morning according to whether the employee used public transportation or drove a car.

Public Transportation

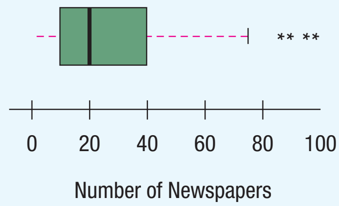
23	25	25	30	31	31	32	33	35	36
37	42								

Private

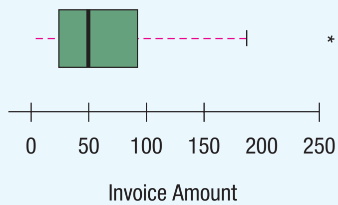
32	32	33	34	37	37	38	38	38	39
40	44								

a. Find the median and the first and third quartiles for the time it took employees using public transportation. Develop a box plot for the information.
b. Find the median and the first and third quartiles for the time it took employees who drove their own vehicle. Develop a box plot for the information.
c. Compare the times of the two groups.

34. The following box plot shows the number of daily newspapers published in each state and the District of Columbia. Write a brief report summarizing the number published. Be sure to include information on the values of the first and third quartiles, the median, and whether there is any skewness. If there are any outliers, estimate their value.



35. Walter Gogel Company is an industrial supplier of fasteners, tools, and springs. The amounts of its invoices vary widely, from less than $20.00 to more than $400.00. During the month of January the company sent out 80 invoices. Here is a box plot of these invoices. Write a brief report summarizing the invoice amounts. Be sure to include information on the values of the first and third quartiles, the median, and whether there is any skewness. If there are any outliers, approximate the value of these invoices.



36. **FILE** The American Society of PeriAnesthesia Nurses (ASPAN; **www.aspan.org**) is a national organization serving nurses practicing in ambulatory surgery, preanesthesia, and postanesthesia care. The organization consists of the 40 components listed below.

State/Region	Membership	State/Region	Membership
Alabama	95	New Jersey, Bermuda	517
Arizona	399	Alaska, Idaho, Montana, Oregon, Washington	708
Maryland, Delaware, DC	531	New York	891
Connecticut	239	Ohio	708
Florida	631	Oklahoma	171
Georgia	384	Arkansas	68
Hawaii	73	California	1,165
Illinois	562	New Mexico	79
Indiana	270	Pennsylvania	575
Iowa	117	Rhode Island	53
Kentucky	197	Colorado	409
Louisiana	258	South Carolina	237
Michigan	411	Texas	1,026
Massachusetts	480	Tennessee	167
Maine	97	Utah	67
Minnesota, Dakotas	289	Virginia	414
Missouri, Kansas	282	Vermont, New Hampshire	144
Mississippi	90	Wisconsin	311
Nebraska	115	West Virginia	62
North Carolina	542		
Nevada	106		

Use statistical software to answer the following questions.

a. Find the mean, median, and standard deviation of the number of members per component.

b. Find the coefficient of skewness, using the software. What do you conclude about the shape of the distribution of component size?
c. Compute the first and third quartiles using formula (4–1).
d. Develop a box plot. Are there any outliers? Which components are outliers? What are the limits for outliers?

37. FILE McGivern Jewelers is located in the Levis Square Mall just south of Toledo, Ohio. Recently it posted an advertisement on a social media site reporting the shape, size, price, and cut grade for 33 of its diamonds currently in stock. The information is reported below.

Shape	Size (carats)	Price	Cut Grade	Shape	Size (carats)	Price	Cut Grade
Princess	5.03	$44,312	Ideal cut	Round	0.77	$2,828	Ultra ideal cut
Round	2.35	20,413	Premium cut	Oval	0.76	3,808	Premium cut
Round	2.03	13,080	Ideal cut	Princess	0.71	2,327	Premium cut
Round	1.56	13,925	Ideal cut	Marquise	0.71	2,732	Good cut
Round	1.21	7,382	Ultra ideal cut	Round	0.70	1,915	Premium cut
Round	1.21	5,154	Average cut	Round	0.66	1,885	Premium cut
Round	1.19	5,339	Premium cut	Round	0.62	1,397	Good cut
Emerald	1.16	5,161	Ideal cut	Round	0.52	2,555	Premium cut
Round	1.08	8,775	Ultra ideal cut	Princess	0.51	1,337	Ideal cut
Round	1.02	4,282	Premium cut	Round	0.51	1,558	Premium cut
Round	1.02	6,943	Ideal cut	Round	0.45	1,191	Premium cut
Marquise	1.01	7,038	Good cut	Princess	0.44	1,319	Average cut
Princess	1.00	4,868	Premium cut	Marquise	0.44	1,319	Premium cut
Round	0.91	5,106	Premium cut	Round	0.40	1,133	Premium cut
Round	0.90	3,921	Good cut	Round	0.35	1,354	Good cut
Round	0.90	3,733	Premium cut	Round	0.32	896	Premium cut
Round	0.84	2,621	Premium cut				

a. Develop a box plot of the variable price and comment on the result. Are there any outliers? What is the median price? What are the values of the first and the third quartiles?
b. Develop a box plot of the variable size and comment on the result. Are there any outliers? What is the median price? What are the values of the first and the third quartiles?
c. Develop a scatter diagram between the variables price and size. Be sure to put price on the vertical axis and size on the horizontal axis. Does there seem to be an association between the two variables? Is the association direct or indirect? Does any point seem to be different from the others?
d. Develop a contingency table for the variables shape and cut grade. What is the most common cut grade? What is the most common shape? What is the most common combination of cut grade and shape?

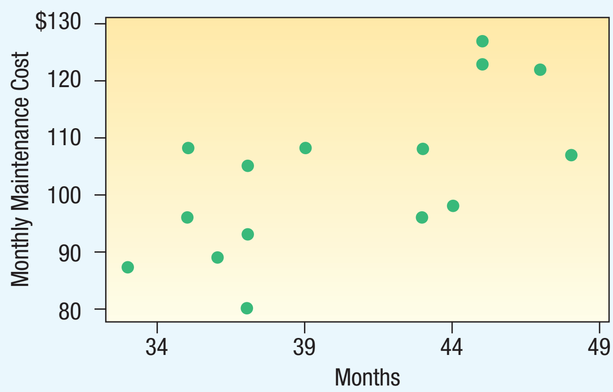
38. FILE Listed below is the amount of commissions earned last month for the eight members of the sales staff at Best Electronics. Calculate the coefficient of skewness using both methods. Hint: Use of a spreadsheet will expedite the calculations.

980.9	1,036.5	1,099.5	1,153.9	1,409.0	1,456.4	1,718.4	1,721.2

39. FILE Listed below is the number of car thefts in a large city over the last week. Calculate the coefficient of skewness using both methods. Hint: Use of a spreadsheet will expedite the calculations.

3	12	13	7	8	3	8

40. The manager of Information Services at Wilkin Investigations, a private investigation firm, is studying the relationship between the age (in months) of a combination printer, copier, and fax machine and its monthly maintenance cost. For a sample of 15 machines, the manager developed the following chart. What can the manager conclude about the relationship between the variables?



41. **FILE** An auto insurance company reported the following information regarding the age of a driver and the number of accidents reported last year. Develop a scatter diagram for the data and write a brief summary.

Age	Accidents	Age	Accidents
16	4	23	0
24	2	27	1
18	5	32	1
17	4	22	3

42. Wendy's offers eight different condiments (mustard, catsup, onion, mayonnaise, pickle, lettuce, tomato, and relish) on hamburgers. A store manager collected the following information on the number of condiments ordered and the age group of the customer. What can you conclude regarding the information? Who tends to order the most or least number of condiments?

	Age			
Number of Condiments	Under 18	18 up to 40	40 up to 60	60 or older
0	12	18	24	52
1	21	76	50	30
2	39	52	40	12
3 or more	71	87	47	28

43. Here is a table showing the number of employed and unemployed workers 20 years or older by gender in the United States.

	Number of Workers (000)	
Gender	Employed	Unemployed
Men	70,415	4,209
Women	61,402	3,314

a. How many workers were studied?
b. What percent of the workers were unemployed?
c. Compare the percent unemployed for the men and the women.

DATA ANALYTICS

44. FILE Refer to the North Valley real estate data recorded on homes sold during the last year. Prepare a report on the selling prices of the homes based on the answers to the following questions.

a. Compute the minimum, maximum, median, and the first and the third quartiles of price. Create a box plot. Comment on the distribution of home prices.

b. Develop a scatter diagram with price on the vertical axis and the size of the home on the horizontal. Is there a relationship between these variables? Is the relationship direct or indirect?

c. For homes without a pool, develop a scatter diagram with price on the vertical axis and the size of the home on the horizontal. Do the same for homes with a pool. How do the relationships between price and size for homes without a pool and homes with a pool compare?

45. FILE Refer to the Baseball 2016 data that report information on the 30 Major League Baseball teams for the 2016 season.

a. In the data set, the year opened, is the first year of operation for that stadium. For each team, use this variable to create a new variable, stadium age, by subtracting the value of the variable, year opened, from the current year. Develop a box plot with the new variable, age. Are there any outliers? If so, which of the stadiums are outliers?

b. Using the variable, salary, create a box plot. Are there any outliers? Compute the quartiles using formula (4–1). Write a brief summary of your analysis.

c. Draw a scatter diagram with the variable, wins, on the vertical axis and salary on the horizontal axis. What are your conclusions?

d. Using the variable, wins, draw a dot plot. What can you conclude from this plot?

46. FILE Refer to the Lincolnville School District bus data.

a. Referring to the maintenance cost variable, develop a box plot. What are the minimum, first quartile, median, third quartile, and maximum values? Are there any outliers?

b. Using the median maintenance cost, develop a contingency table with bus manufacturer as one variable and whether the maintenance cost was above or below the median as the other variable. What are your conclusions?

A REVIEW OF CHAPTERS 1–4

This section is a review of the major concepts and terms introduced in Chapters 1–4. Chapter 1 began by describing the meaning and purpose of statistics. Next we described the different types of variables and the four levels of measurement. Chapter 2 was concerned with describing a set of observations by organizing it into a frequency distribution and then portraying the frequency distribution as a histogram or a frequency polygon. Chapter 3 began by describing measures of location, such as the mean, weighted mean, median, geometric mean, and mode. This chapter also included measures of dispersion, or spread. Discussed in this section were the range, variance, and standard deviation. Chapter 4 included several graphing techniques such as dot plots, box plots, and scatter diagrams. We also discussed the coefficient of skewness, which reports the lack of symmetry in a set of data.

Throughout this section we stressed the importance of statistical software, such as Excel and Minitab. Many computer outputs in these chapters demonstrated how quickly and effectively a large data set can be organized into a frequency distribution, several of the measures of location or measures of variation calculated, and the information presented in graphical form.

PROBLEMS

1. FILE The duration in minutes of a sample of 50 power outages last year in the state of South Carolina is listed below.

124	14	150	289	52	156	203	82	27	248
39	52	103	58	136	249	110	298	251	157
186	107	142	185	75	202	119	219	156	78
116	152	206	117	52	299	58	153	219	148
145	187	165	147	158	146	185	186	149	140

Use a statistical software package such as Excel or Minitab to help answer the following questions.

a. Determine the mean, median, and standard deviation.
b. Determine the first and third quartiles.
c. Develop a box plot. Are there any outliers? Do the amounts follow a symmetric distribution or are they skewed? Justify your answer.
d. Organize the distribution of funds into a frequency distribution.
e. Write a brief summary of the results in parts a to d.

2. FILE Listed below are the 45 U.S. presidents and their age as they began their terms in office.

Number	Name	Age	Number	Name	Age
1	Washington	57	24	Cleveland	55
2	J. Adams	61	25	McKinley	54
3	Jefferson	57	26	T. Roosevelt	42
4	Madison	57	27	Taft	51
5	Monroe	58	28	Wilson	56
6	J. Q. Adams	57	29	Harding	55
7	Jackson	61	30	Coolidge	51
8	Van Buren	54	31	Hoover	54
9	W. H. Harrison	68	32	F. D. Roosevelt	51
10	Tyler	51	33	Truman	60
11	Polk	49	34	Eisenhower	62
12	Taylor	64	35	Kennedy	43
13	Fillmore	50	36	L. B. Johnson	55
14	Pierce	48	37	Nixon	56
15	Buchanan	65	38	Ford	61
16	Lincoln	52	39	Carter	52
17	A. Johnson	56	40	Reagan	69
18	Grant	46	41	G. H. W. Bush	64
19	Hayes	54	42	Clinton	46
20	Garfield	49	43	G. W. Bush	54
21	Arthur	50	44	Obama	47
22	Cleveland	47	45	Trump	70
23	B. Harrison	55			

Use a statistical software package such as Excel or Minitab to help answer the following questions.

a. Determine the mean, median, and standard deviation.
b. Determine the first and third quartiles.
c. Develop a box plot. Are there any outliers? Do the amounts follow a symmetric distribution or are they skewed? Justify your answer.
d. Organize the distribution of ages into a frequency distribution.
e. Write a brief summary of the results in parts a to d.

3. FILE Listed below is the 2014 median household income for the 50 states and the District of Columbia. https://www.census.gov/hhes/www/income/data/historical/household/

State	Amount	State	Amount
Alabama	42,278	Montana	51,102
Alaska	67,629	Nebraska	56,870
Arizona	49,254	Nevada	49,875
Arkansas	44,922	New Hampshire	73,397
California	60,487	New Jersey	65,243
Colorado	60,940	New Mexico	46,686
Connecticut	70,161	New York	54,310
Delaware	57,522	North Carolina	46,784
D.C.	68,277	North Dakota	60,730
Florida	46,140	Ohio	49,644
Georgia	49,555	Oklahoma	47,199
Hawaii	71,223	Oregon	58,875
Idaho	53,438	Pennsylvania	55,173
Illinois	54,916	Rhode Island	58,633
Indiana	48,060	South Carolina	44,929
Iowa	57,810	South Dakota	53,053
Kansas	53,444	Tennessee	43,716
Kentucky	42,786	Texas	53,875
Louisiana	42,406	Utah	63,383
Maine	51,710	Vermont	60,708
Maryland	76,165	Virginia	66,155
Massachusetts	63,151	Washington	59,068
Michigan	52,005	West Virginia	39,552
Minnesota	67,244	Wisconsin	58,080
Mississippi	35,521	Wyoming	55,690
Missouri	56,630		

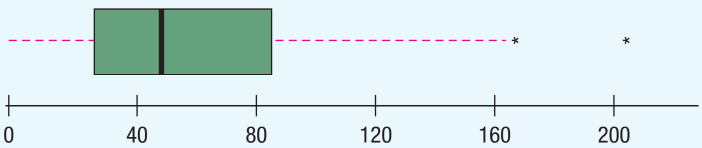
Use a statistical software package such as Excel or Minitab to help answer the following questions.

a. Determine the mean, median, and standard deviation.
b. Determine the first and third quartiles.
c. Develop a box plot. Are there any outliers? Do the amounts follow a symmetric distribution or are they skewed? Justify your answer.
d. Organize the distribution of funds into a frequency distribution.
e. Write a brief summary of the results in parts a to d.

4. A sample of 12 homes sold last week in St. Paul, Minnesota, revealed the following information. Draw a scatter diagram. Can we conclude that, as the size of the home (reported below in thousands of square feet) increases, the selling price (reported in $ thousands) also increases?

Home Size (thousands of square feet)	Selling Price ($ thousands)	Home Size (thousands of square feet)	Selling Price ($ thousands)
1.4	100	1.3	110
1.3	110	0.8	85
1.2	105	1.2	105
1.1	120	0.9	75
1.4	80	1.1	70
1.0	105	1.1	95

5. Refer to the following diagram.



a. What is the graph called?
b. What are the median, and first and third quartile values?
c. Is the distribution positively skewed? Tell how you know.
d. Are there any outliers? If yes, estimate these values.
e. Can you determine the number of observations in the study?

CASES

A. Century National Bank

The following case will appear in subsequent review sections. Assume that you work in the Planning Department of the Century National Bank and report to Ms. Lamberg. You will need to do some data analysis and prepare a short written report. Remember, Mr. Selig is the president of the bank, so you will want to ensure that your report is complete and accurate. A copy of the data appears in Appendix A.6.

Century National Bank has offices in several cities in the Midwest and the southeastern part of the United States. Mr. Dan Selig, president and CEO, would like to know the characteristics of his checking account customers. What is the balance of a typical customer?

How many other bank services do the checking account customers use? Do the customers use the ATM service and, if so, how often? What about debit cards? Who uses them, and how often are they used?

To better understand the customers, Mr. Selig asked Ms. Wendy Lamberg, director of planning, to select a sample of customers and prepare a report. To begin, she has appointed a team from her staff. You are the head of the team and responsible for preparing the report. You select a random sample of 60 customers. In addition to the balance in each account at the end of last month, you determine (1) the number of ATM (automatic teller machine) transactions in the last month; (2) the number of other bank services (a savings account, a certificate of deposit, etc.) the customer uses; (3) whether the customer has a debit card (this is a bank service in which charges are made directly to the customer's account); and (4) whether or not interest is paid on the checking account. The sample includes customers from the branches in Cincinnati, Ohio; Atlanta, Georgia; Louisville, Kentucky; and Erie, Pennsylvania.

1. Develop a graph or table that portrays the checking balances. What is the balance of a typical customer? Do many customers have more than $2,000 in their accounts? Does it appear that there is a difference in the distribution of the accounts among the four branches? Around what value do the account balances tend to cluster?

2. Determine the mean and median of the checking account balances. Compare the mean and the median balances for the four branches. Is there a difference among the branches? Be sure to explain the difference between the mean and the median in your report.

3. Determine the range and the standard deviation of the checking account balances. What do the first and third quartiles show? Determine the coefficient of skewness and indicate what it shows. Because Mr. Selig does not deal with statistics daily, include a brief description and interpretation of the standard deviation and other measures.

B. Wildcat Plumbing Supply Inc.: Do We Have Gender Differences?

Wildcat Plumbing Supply has served the plumbing needs of Southwest Arizona for more than 40 years. The company was founded by Mr. Terrence St. Julian and is run today by his son Cory. The company has grown from a handful of employees to more than 500 today. Cory is concerned about several positions within the company where he has men and women doing essentially the same job but at different pay. To investigate, he collected the information below. Suppose you are a student intern in the Accounting Department and have been given the task to write a report summarizing the situation.

Yearly Salary ($000)	Women	Men
Less than 30	2	0
30 up to 40	3	1
40 up to 50	17	4
50 up to 60	17	24
60 up to 70	8	21
70 up to 80	3	7
80 or more	0	3

To kick off the project, Mr. Cory St. Julian held a meeting with his staff and you were invited. At this meeting, it was suggested that you calculate several measures of location, create charts or draw graphs such as a cumulative frequency distribution, and determine the quartiles for both men and women. Develop the charts and write the report summarizing the yearly salaries of employees at Wildcat Plumbing Supply. Does it appear that there are pay differences based on gender?

C. Kimble Products: Is There a Difference In the Commissions?

At the January national sales meeting, the CEO of Kimble Products was questioned extensively regarding the company policy for paying commissions to its sales representatives. The company sells sporting goods to two major markets. There are 40 sales representatives who call directly on large-volume customers, such as the athletic departments at major colleges and universities and professional sports franchises. There are 30 sales representatives who represent the company to retail stores located in shopping malls and large discounters such as Kmart and Target.

Upon his return to corporate headquarters, the CEO asked the sales manager for a report comparing the commissions earned last year by the two parts of the sales team. The information is reported below. Write a brief report. Would you conclude that there is a difference? Be sure to include information in the report on both the central tendency and dispersion of the two groups.

Commissions Earned by Sales Representatives Calling on Athletic Departments ($)

354	87	1,676	1,187	69	3,202	680	39	1,683	1,106
883	3,140	299	2,197	175	159	1,105	434	615	149
1,168	278	579	7	357	252	1,602	2,321	4	392
416	427	1,738	526	13	1,604	249	557	635	527

Commissions Earned by Sales Representatives Calling on Large Retailers ($)

1,116	681	1,294	12	754	1,206	1,448	870	944	1,255
1,213	1,291	719	934	1,313	1,083	899	850	886	1,556
886	1,315	1,858	1,262	1,338	1,066	807	1,244	758	918

PRACTICE TEST

There is a practice test at the end of each review section. The tests are in two parts. The first part contains several objective questions, usually in a fill-in-the-blank format. The second part is problems. In most cases, it should take 30 to 45 minutes to complete the test. The problems require a calculator. Check the answers in the Answer Section in the back of the book.

Part 1—Objective

1. The science of collecting, organizing, presenting, analyzing, and interpreting data to assist in making effective decisions is called ________. **1.** ________
2. Methods of organizing, summarizing, and presenting data in an informative way are called ________. **2.** ________
3. The entire set of individuals or objects of interest or the measurements obtained from all individuals or objects of interest are called the ________. **3.** ________
4. List the two types of variables. **4.** ________
5. The number of bedrooms in a house is an example of a ________. (discrete variable, continuous variable, qualitative variable—pick one) **5.** ________
6. The jersey numbers of Major League Baseball players are an example of what level of measurement? **6.** ________
7. The classification of students by eye color is an example of what level of measurement? **7.** ________
8. The sum of the differences between each value and the mean is always equal to what value? **8.** ________
9. A set of data contained 70 observations. How many classes would the 2^k method suggest to construct a frequency distribution? **9.** ________
10. What percent of the values in a data set are always larger than the median? **10.** ________
11. The square of the standard deviation is the ________. **11.** ________
12. The standard deviation assumes a negative value when ________. (all the values are negative, at least half the values are negative, or never—pick one.) **12.** ________
13. Which of the following is least affected by an outlier? (mean, median, or range—pick one) **13.** ________

Part 2—Problems

1. The Russell 2000 index of stock prices increased by the following amounts over the last 3 years.

18%	4%	2%

What is the geometric mean increase for the 3 years? ________

2. The information below refers to the selling prices ($000) of homes sold in Warren, Pennsylvania, during 2016.

Selling Price ($000)	Frequency
120.0 up to 150.0	4
150.0 up to 180.0	18
180.0 up to 210.0	30
210.0 up to 240.0	20
240.0 up to 270.0	17
270.0 up to 300.0	10
300.0 up to 330.0	6

a. What is the class interval? __________
b. How many homes were sold in 2016? __________
c. How many homes sold for less than $210,000? __________
d. What is the relative frequency of the 210 up to 240 class? __________
e. What is the midpoint of the 150 up to 180 class? __________
f. The selling prices range between what two amounts? __________

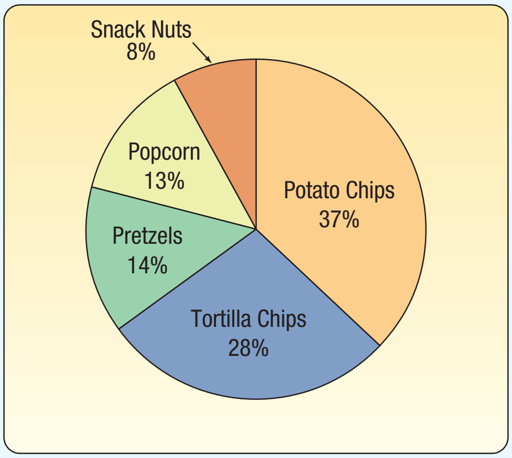
3. A sample of eight college students revealed they owned the following number of CDs.

52	76	64	79	80	74	66	69

a. What is the mean number of CDs owned? __________
b. What is the median number of CDs owned? __________
c. What is the 40th percentile? __________
d. What is the range of the number of CDs owned? __________
e. What is the standard deviation of the number of CDs owned? __________

4. An investor purchased 200 shares of the Blair Company for $36 each in July of 2013, 300 shares at $40 each in September 2015, and 500 shares at $50 each in January 2016. What is the investor's weighted mean price per share? __________

5. During the 50th Super Bowl, 30 million pounds of snack food were eaten. The chart below depicts this information.



a. What is the name given to this graph? __________
b. Estimate, in millions of pounds, the amount of potato chips eaten during the game. __________
c. Estimate the relationship of potato chips to popcorn. (twice as much, half as much, three times, none of these—pick one) __________
d. What percent of the total do potato chips and tortilla chips comprise? __________